

Degree Project in Technology

First Cycle, 15 Credits

Collaboration Within the Quadruple Helix Framework

A Study on How Cultural Differences Among Sectors Influence Collaboration

NELLY BENGTSSON & NOOMI CARLE

KTH Industriell teknik och management

Examensarbete Integrerad produktutveckling grundnivå, 15 hp, kurskod MF131x

Samarbete inom Quadruple Helix ramverket

En studie om hur kulturella skillnader mellan sektorer påverkar samarbetet

Nelly Bengtsson
Noomi Carle

Examinator	Handledare
Gunilla Ölundh Sandström	Magnus Eneberg och Gunilla Ölundh Sandström

Sammanfattning

Quadruple Helix är en innovationsmodell som knyter samman akademi, privata sektorn, myndigheter och civilsamhället. Denna innovationsmodell är ett ramverk för samskapande som syftar till att främja synergi och innovation. Detta går i linje med FN:s Agenda 2030, och särskilt mål nummer 17 som uppmuntrar till samarbete, partnerskap och vikten av kunskapsdelning.

Syftet med denna studie var att undersöka upplevda styrkor och utmaningar i sammarbetet inom en Quadruple Helix-kontext, med särskilt fokus på kommunikation och hur skillnader mellan sektorer kan överbryggas.

Studien genomfördes som en kvalitativ studie där semistrukturerade intervjuer användes som den primära metoden för insamling av data. Intervjuerna genomfördes med sju respondenter som alla hade tidigare erfarenhet av att arbeta inom en Quadruple Helix-kontext.

Resultatet visade att när samarbetet fungerar effektivt skapas förutsättningar för synergi och innovation, tack vare variationen av erfarenheter och expertisområden som finns representerade. Samtidigt identifierades flera utmaningar i samarbetet, såsom skillnader i värderingar, arbetsmetoder och ett utpräglat stuprörstänkande.

Resultatet visade även att kommunikation var en viktig faktor i samarbetet. Användningen av akronymer och facktermer kan potentiellt leda till missförstånd eller exkludering. Vissa begrepp kräver mer varsam hantering än andra, och en brist på medvetenhet eller lyhördhet kan leda till oavsiktlig spänning eller konflikt.

Nyckelord:
Quadruple Helix, QH, samarbete, kommunikation, interkulturell kompetens, innovationsmodell, utvecklingsprocess

KTH Industrial Engineering and Management

Degree project in Integrated Product Development
First cycle, 15 cr, course code MF131x

Collaboration Within the Quadruple Helix Framework

A Study on How Cultural Differences Among Sectors Influence Collaboration

Nelly Bengtsson
Noomi Carle

Examiner	Supervisor
Gunilla Ölundh Sandström	Magnus Eneberg and Gunilla Ölundh Sandström

Abstract

Quadruple Helix is an innovative model connecting the Academia, Government, Private Sector and Civil Society through a co-creating framework aimed to foster synergy and innovation. This approach aligns with the United Nation's Agenda 2030, especially goal number 17, which encourages collaboration, partnerships and the importance of sharing knowledge.

The aim of this research was to explore the perceived strengths and challenges of collaboration in a Quadruple Helix setting, with a particular focus on communication and how the differences between sectors can be bridged.

The qualitative study was conducted using semi-structured interviews as the primary method for data collection. The interviews involved seven participants with experience working in a Quadruple Helix setting.

From the study it was shown that if the collaboration is working effectively, the diversity of expertise and experience creates momentum for both synergy and innovation. While there are challenges of the collaboration such as differences in values, work methods and patterns of silo-thinking.

The findings also showed that communication is an important factor in the collaboration. The use of acronyms and internal language can potentially lead to misunderstandings or exclusion. Some terms require more careful and context-sensitive use than others, and a lack of awareness or sensitivity may result in unintended tension or conflict.

Keywords:
Quadruple Helix, QH, cross-sectoral collaboration, communication, intercultural competence, innovation framework, development processes

Acknowledgements

This bachelor's thesis was carried out at KTH Royal Institute of Technology as the final part of our Bachelor's degree in Design and Product Development.

We would like to thank our course instructors, Gunilla Ölundh Sandström and Magnus Eneberg, for their guidance throughout the process. A special thanks to our supervisor Magnus for his valuable feedback and encouragement.

We also want to thank all interview participants for generously sharing their time, experiences, and insights.

Stockholm, June 2025

Nelly Bengtsson & Noomi Carle

Contents

1 Introduction **1**
- 1.1 Background . . . 1
- 1.2 Aim . . . 1
- 1.3 Research Questions . . . 2
- 1.4 Clarifications . . . 2
- 1.5 Sustainable Development Goals . . . 2

2 Theoretical Framework **3**
- 2.1 Development Processes . . . 3
- 2.2 Quadruple Helix . . . 4
- 2.3 Collaboration in Teams . . . 5
- 2.4 Organizational Culture . . . 6
- 2.5 Intercultural Competence . . . 7

3 Method **8**
- 3.1 Literature Review . . . 8
- 3.2 Qualitative Study . . . 8
- 3.3 Selection of Respondents . . . 8
- 3.4 Semi-Structured Interviews . . . 10
- 3.5 Processing the Interviews . . . 11

4 Results **12**
- 4.1 Respondents' View on the Quadruple Helix . . . 12
- 4.2 Differences Between the Sectors . . . 13
- 4.3 Communication . . . 15
- 4.4 Building Bridges . . . 16

5 Analysis and Discussion **18**
- 5.1 Strengths and Challenges of the Collaboration . . . 18
- 5.2 How Communication Affects the Collaboration . . . 19
- 5.3 Bridging Differences Between Sectors . . . 20
- 5.4 Method Discussion . . . 21

6 Conclusions **22**
- 6.1 Recommendations for Actors in the Quadruple Helix . . . 22
- 6.2 Suggestions for Further Studies . . . 23

References **24**

Appendix A - Interview Guide in Swedish **I**

Appendix B - Interview Guide in English **III**

Appendix C - Quotes in Original Language **V**

1 Introduction

This section presents the background, aim, and research questions of the study.

1.1 Background

The world is facing critical challenges related to sustainable development. Agenda 2030 is a plan to address these challenges with 17 Sustainable Development Goals (SDGs) aimed to stimulate global action (United Nations, 2015a) (see Section 1.5).

Goal 17 of Agenda 2030, *Partnership for the Goals*, encourages collaboration and promotes effective partnerships (United Nations, 2015b). One approach to strengthening the kind of partnerships necessary to face critical challenges in the Quadruple Helix framework, which integrates Academia, the Private Sector, Government and Civil Society in innovation processes (Schütz et al., 2019).

Quadruple Helix as an innovation model in development processes is mainly used within the context of a project. Project as a form serves as the most commonly used working method to bring change and development in contemporary contexts (Scott, 2023).

A development process involves creation, improvements or implementation of new products, services or methods (Ulrich & Eppinger, 2014). A successful development process is often built on a team with highly effective knowledge exchange, high capacity of innovation and collaboration between various individuals and groups (Engelsberger et al., 2023; Park et al., 2021). In Engelsberger et al. (2023) it is further argued that a highly effective team with strong collaborative ability is characterized by a high level of social interaction, trust and shared commitment for the stated goals.

Although the collaboration between sectors is very important for creating sustainable impact in line with the SDGs it does not always happen without challenges. In Dobers and Stier (2018) it is indicated that differences in culture, mentality and institutional logics between the sectors can lead to obstacles for the collaboration. Even if a project has a clear goal, the different sectors may have conflicting interests, approaches and strategies to achieve it (Nguyen & Marques, 2022).

1.2 Aim

The aim of this study is to explore the perceived strengths and challenges of collaboration within a Quadruple Helix setting, with a particular focus on communication and how differences between sectors can be bridged.

1.3 Research Questions

- What perceived challenges and strengths do people with experience in Quadruple Helix projects see in cross-sector collaboration?
- How does communication affect collaboration across sectors?
- How can differences between sectors be bridged to improve collaboration?

1.4 Clarifications

When reviewing the literature on the Quadruple Helix, it is clear that the same sector is often described using different terms. *Academia* is also called *University* or *Science*. The *Private Sector* may be referred to as Industry, *Firms* or *Business*. *Government* is described as *Policymakers* or the *Public Sector*. *Civil Society* is sometimes called *Society*, *Users*, *Citizens* or *Consumers*. There are likely many more terms and variations used in other sources that are not included here.

In this report, *Civil Society* is represented exclusively by Non-Governmental Organizations (NGOs).

Throughout the text, we use the following terms to describe the four sectors: *Academia, Government, Private Sector* and *Civil Society*.

1.5 Sustainable Development Goals

This degree project addresses Goal 17, *Partnership for the Goals*. This Goal encourages collaboration and promotes effective partnerships for reaching the Agenda 2030. It highlights global partnerships and the importance of sharing knowledge, expertise, technology and financial resources to ensure sustainable development (United Nations Children's Fund (UNICEF), n.d.).

2 Theoretical Framework

This section presents the theoretical framework for understanding development processes, and the Quadruple Helix model. It includes factors that shape co-creation across sectors, such as collaboration in teams, organizational culture, and intercultural competence.

2.1 Development Processes

Understanding how innovations emerge and takes shape in collaborative settings requires a foundation in development process theory. A development process involves creation, improvements or implementation of new products, services or methods (Ulrich & Eppinger, 2014).

Although a process is often seen as a sequence of steps used to complete a task, the concept is broad and the meaning can vary depending on the context. In design and development contexts, a process refers to the act of creating and developing a product (Wynn & Clarkson, 2024). In this case, where innovation occurs through multi-sector collaborations, the process appears to be more iterative and socially embedded.

In Wynn and Clarkson (2024), it's further suggested that innovation is a process closely linked to development processes. One explanation of the term is when using an idea, invention or method to improve or create a new product or services in a way that meets a clear need and creates value (Castaneda & Cuellar, 2020). This is important in both economic growth and social development (Parveen et al., 2015).

2.1.1 Cross-Sectoral Innovation

To understand the essential components for a cross-sectoral initiative to work, knowledge sharing is a term that becomes relevant almost immediately. It has been argued in Castaneda and Cuellar (2020) that knowledge sharing fosters innovation as it includes sharing experiences, skills, and insights between people.

An earlier model used to foster innovation in cross-sectoral contexts is the Triple Helix model, which describes how innovation is created through the interaction between Government, Academia and the Private Sector. While each of the three sectors have their distinct structures and roles, the model emphasizes the overlaps of knowledge areas as well as the influence coming from all involved parties. The model was first introduced in Leydesdorff and Etzkowitz (1998).

Although Triple Helix has been used as a tool to understand synergies in cross-collaborative contexts, the model has also been criticized for its lack of

involvement of end users, i.e. the people who use the products or services developed (Van Horne & Dutot, 2017). In response to this gap, a fourth helix was proposed: Civil Society (Carayannis & Campbell, 2012). The revised model of Triple Helix has now become Quadruple Helix, enhancing the relevance of the model by incorporating the perspectives and needs of the broader public.

2.2 Quadruple Helix

Introduced in Carayannis and Campbell (2012), the fourth sector of Quadruple Helix, i.e. Civil Society, brings valuable user perspectives into the co-creating process, which was previously dominated by scientific and technical input from the other three sectors. Under the helix of Civil Society lies various civil society organizations.

Typical for civil society organizations is their close work with local communities. The insights gained by the civil society organizations make them key actors in fostering citizen involvement and public engagement, as well as important contributors to research and innovation (Dobers & Stier, 2018). A positive attitude towards the addition of Civil Society is expressed in González-Martinez et al. (2021), where it is argued that Civil Society can further help businesses understand the needs of market and reduce risks in product development. However, despite the relevance of including civil society organizations, they often face challenges related to limited funding (Dobers & Stier, 2018).

The four sectors have their own unique resources and competencies which, in the best case, can be shared across sectors to enhance innovation and knowledge sharing (Parveen et al., 2015). Figure 2.1 shows the Quadruple helix model and an overview of the sectors.

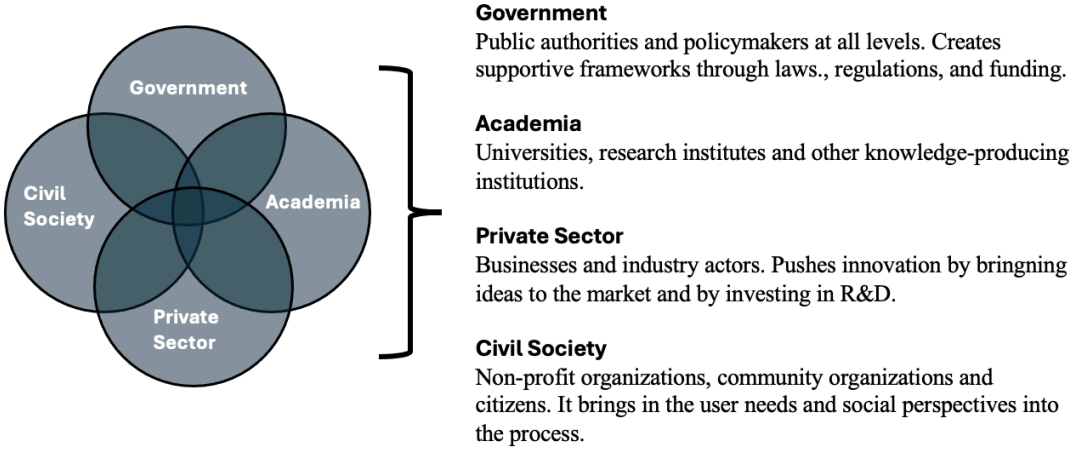


Figure 2.1: Quadruple Helix model and overview of the sectors. Own illustration adapted from Lindberg et al. (2014, p. 109).

2.3 Collaboration in Teams

Collaboration is often seen as important for team creativity and a well-functioning collaboration is also a key to a successful development process. As described in Engelsberger et al. (2023), a team with strong collaboration involves high levels of social exchange, trust, and shared commitment to goals. However, collaboration is not always easy to achieve in practice.

When people come from different sectors or have different expertise, there may be differences in use of terminology, work methods, or professional values and these differences can create communication barriers and limit the team's ability to make use of its full potential (Engelsberger et al., 2023).

2.3.1 Challenges in Quadruple Helix Collaboration

Although collaboration across sectors is central to the Quadruple Helix model, it presents several challenges. Differences in cultures, goals, timelines, roles, institutional logics, and incentives can make collaboration difficult (Dobers & Stier, 2018). In Dobers and Stier (2018) it is argued that having a shared goal is essential to achieve the desired results.

Communication is important, including the importance of adjusting the communication based on the target group (Dobers & Stier, 2018). Other challenges in Quadruple Helix collaboration includes factors such as differences in time horizon management and the adaption of theory to practice (Van Horne & Dutot, 2017).

Pointed out in Dobers and Stier (2018) is the role of *intermediaries*, which are people who understand multiple sectors and can ”translate” between them to improve mutual understanding (Dobers & Stier, 2018).

2.3.2 Cross-Functional Teams

To better understand collaboration in Quadruple Helix projects, the theory of cross-functional teams is used to explore how team dynamics and communication affect the ability to work across different sectors.

In development processes, teams are often cross-functional. This means they include people with different skills and areas of expertise working together. These diverse perspectives can improve problem solving and support innovation (Daspit et al., 2013). However, such diversity also introduces risks. If the team cannot create a supportive environment for collaboration, its diversity may become a source of tension instead of strength, and the potential value of their diverse expertise may be lost.

In Daspit et al. (2013) it is suggested that learning communication techniques

can help team members see that not all conflicts are inherently negative. One example is learning to distinguish between cognitive conflict, which is related to tasks and can sometimes be productive, and affective conflict, which is personal and related to relationships and often harmful. This awareness can help teams manage tensions and engage in more constructive forms of conflict.

Leadership in cross-functional does not come from one fixed leader but is rather shared and dynamic (Daspit et al., 2013). It depends on the ability of the team members to collaborate and the leadership distributes among multiple members.

2.4 Organizational Culture

Culture is a multifaceted term that lacks a clear, singular definition. A general definition of the term usually refers to ”prescribed ethnic, national and international entities” (Holliday, 1999). Other definitions focus more on what happens inside a group. A definition in Birukou et al. (2013) emphasizes other aspects of the culture, such as ”norms, values, shared meanings” within a group. It is further argued that culture should not be viewed solely from a geographical or religious point of view, as factors such as globalization have made this previous distinction less meaningful (Birukou et al., 2013).

In response to this, the concept of *small culture* was introduced (Holliday, 1999). Small culture describes self-organized groups brought together by shared beliefs or understandings, rather than being categorized according to national or religious prescriptions. The notion of small culture is important when building a collaborative team with different backgrounds since it suggests that the co-creating framework is more shaped by shared values rather than formal structures.

When considering how organizations contribute to the formation of social groups, the concept of organizational culture becomes particularly relevant. Organizational culture can be described as ”the values and assumptions shared by members of an organization” (Kava et al., 2018). In a team made up of people from different sectors in the Quadruple Helix, this means building a shared culture within the group. In Parveen et al. (2015) it is highlighted that there is an intimate correlation between organizational culture and the quality of the co-creation carried out through the framework of Quadruple Helix. In Parveen et al. (2015), Schein’s (1992) definition of organizational culture is preferred: ”the values and beliefs that provide norms of expected behaviors that employees might follow”.

2.5 Intercultural Competence

Intercultural competence is a concept that is often regarded ambiguous. There is no universally accepted definition of the term, and numerous attempts have been made to capture and define it. Intercultural competence is in Hammer et al. (2003) described as ”the ability to think and act in interculturally appropriate ways”, while it in Lloyd and Härtel (2010) is defined as ”a set of skills, knowledge and attitudes that are used when interacting with culturally diverse team members”. Although it seems difficult to capture the essence of the concept in its entirety, it is suggested in Leung et al. (2014) that there is still a common understanding, which refers to an individual’s ability to collaborate effectively with representatives from diverse cultural backgrounds. Therefore, intercultural competence may become an important skill of the project members of a Quadruple Helix project.

Given the varying representation of national cultures, corporate cultures, social class, religious status, political views and more, it is argued in Popescu et al. (2014) that such variation is closely linked to the emergence of misunderstandings and conflicts. Even when goodwill is present, the authors note that implicit norms can still cause tension within the group. By recognizing these differences and understanding their underlying causes, intercultural competence can serve as an effective tool for preventing conflict, as it equips the involving parties with the ability to communicate effectively and bridge differences (Lloyd & Härtel, 2010).

3 Method

This section presents the method used in this study. It describes how the theoretical framework was built, how the data was collected, and how the respondents were selected.

3.1 Literature Review

A literature review was conducted to build a theoretical framework and examine the current state of knowledge. To find research articles, KTH Primo and Google Scholar were used as relevant databases. ResearchGate was also used to locate relevant articles, with DOI numbers used to trace them back to their original sources. A selection of the keywords used is listed below:

quadruple helix, organizational culture, collaboration, innovation, teamwork, intercultural competence, intercultural communication, creativity

3.2 Qualitative Study

Semi-structured interviews were chosen as a suitable method because they allow flexibility to explore new themes and insights during the interviews. Seven respondents from three of the four Quadruple Helix sectors participated in the study. The fourth helix, the government, was not represented since the governmental perspective was considered to be introduced through the Program Leader's contribution. Throughout the study, an AI tool was used to support and improve language use, as well as to translate relevant terms and quotations. In the early stages, the AI tool also helped generate ideas. The tool was used cautiously, serving as a tool to refine existing material.

3.3 Selection of Respondents

The only criterion for selecting participants was prior experience in Quadruple Helix-driven projects. Four out of seven respondents were recommended by an NGO involved in such projects, while the remaining three were identified through snowball sampling – a method in which participants are recommended by previous respondents.

As a result of the initial recommendations from the NGO, the respondents representing Civil Society are all associated with that organization.

Table 3.1 lists all participating respondents.

Table 3.1: Overview of respondents roles, what organization they work in and the sector they represent

Role	Organization	Sector
Researcher/ Consultant	University	Academia
Project Manager	Research Institute	Academia
Senior Advisor Business Sector	Public-Private Partnership	Private Sector
Chief Strategy Officer	Start-Up	Private Sector
Country Director	NGO	Civil Society
Executive Officer	NGO	Civil Society
Program Leader	Partnership Platform	Sector Integrator

3.3.1 Academia

The Researcher/ Consultant works at a university conducting research in health-related technology. The respondent has previous experience working in cross-sectoral projects and Quadruple Helix-driven initiatives, and therefore represents the perspective of Academia in this report.

The Project Manager works at an independent research institute focused on environment and sustainability. The institute operates as a non-profit and is owned by a foundation with a board composed of representatives from both the public and private sectors. With more than twenty years of experience in public advisory work, the private sector, and applied research, the respondent primarily represents the perspective of Academia in this report.

3.3.2 Private Sector

The Senior Advisor Business Sector currently works as an advisor at an NGO and as a Business Developer at a partnership platform. With an engineering background, the respondent has primarily worked with business development and innovation strategy, both in Sweden and internationally. Given the respondent's long experience in the private sector and current role as Senior Advisor Business Sector, the respondent primarily represents the Private Sector in this report.

The Chief Strategy Officer works for a start-up developing electric vehicles in a low-income country. The respondent has previous experience in aid-related work within non-profit organizations, offering insights from two of the sectors represented in the Quadruple Helix framework. Given the current role, the respondent is considered a representative of the Private Sector.

3.3.3 Civil Society

The Country Director works for an NGO that implements programs related to humanitarian aid and development. The respondent has over 20 years of experience working with communities and providing aid through the NGO. In this report, the respondent represents Civil Society.

The Executive Officer works for an NGO that implements programs related to humanitarian aid and development. The respondent has seven years of experience working in the development aid sector and has been involved in multiple initiatives and cross-sectoral projects. In this report, the respondent represents Civil Society.

3.3.4 Sector Integrator

The Program Leader works for a partnership platform focused on cross-sectoral collaboration. The respondent has more than thirty years of experience in the field and has developed and led several initiatives aimed at bridging sectoral divides and promoting sustainable innovation within the framework of the Quadruple Helix model. Therefore, the respondent does not represent a single sector but instead contributes a broad perspective on connecting the sectors, making the term “Sector Integrator” more accurate.

3.4 Semi-Structured Interviews

An interview guide was created in Swedish and then translated into English. Given the format of semi-structured interviews, the interviewer asked follow-up questions throughout the interview. The authors took turns conducting the interviews and taking notes. The person taking notes was also given the opportunity to ask follow-up questions at the end. After the first interview, slight changes were made to improve the quality of the questions. The choice of language depended on whether the respondent spoke Swedish fluently or not. All interviews were recorded and conducted over the course of two weeks. Table 3.2 provides an overview of the interview format.

Table 3.2: Overview of the conducted interviews

Role	Platform	Language	Length
Researcher/ Consultant	Zoom	English	50 minutes
Project Manager	Zoom	Swedish	54 minutes
Senior Advisor Business Sector	Zoom	Swedish	72 minutes
Chief Strategy Officer	Zoom	Swedish	53 minutes
Country Director	Zoom	English	57 minutes
Executive Officer	Zoom	English	57 minutes
Program Leader	Zoom	Swedish	54 minutes

3.5 Processing the Interviews

A first draft of the transcribed material was generated using an AI transcription tool. The same author who conducted the interviews was later responsible for reviewing the recorded material. Once the transcripts were validated, both authors independently coded the material. This process involved reading the transcripts individually, highlighting relevant content, and noting keywords for each paragraph.

Using a digital whiteboard, relevant quotes and text segments were placed on digital post-it notes. Different colors were assigned to each respondent. All keywords were discussed, and themes were identified through collaborative clustering of post-it notes.

In the first round of clustering, several small themes were identified. These later merged into the following four main themes:

- *Respondents' View on the Quadruple Helix*
- *Differences Between the Sectors*
- *Communication*
- *Building Bridges*

As the ongoing analysis of the results progressed, the participating respondents were contacted to review and approve the use of their quotations. This also provided them with the opportunity to add to or clarify their statements.

4 Results

This section presents the results of the research.

4.1 Respondents' View on the Quadruple Helix

The Senior Advisor Business Sector emphasizes the *willingness* to understand others as a key factor in strengthening partnerships. According to the respondent, a crucial mindset for successful collaboration is to shift the mindset away from the individual perspective of what each party *does* or *doesn't do*, and instead focus on shared vision, goals, and what can be accomplished together.

> *"If you don't believe in the model of Quadruple Helix, it is doomed to fail."*
>
> - Senior Advisor Business Sector

The Researcher/ Consultant elaborates on this by emphasizing the importance of adopting a 360-degree perspective. Trying to meet the needs and interests of the involved parties increases engagement and commitment, and thereby making the project more sustainable, the Researcher/ Consultant explains.

> *"If it's innovation coming from the investor, if it's been taken up by civil society or a private sector, it's going to be owned, it's going to be sustainable."*
>
> - Researcher/ Consultant

Looking at Quadruple Helix projects from the perspective of resource allocation, the Chief Strategy Officer raises the question of whether the resources invested truly align with the outcomes produced, given the exhausting climate that the Quadruple Helix can imply. The remaining respondents are still convinced of the reward of the efforts, expressing sympathy with the Executive Officer.

> *"It's very stressful, yes, but if it goes through, the rewarding is quite good. The project is grounded. There's ownership, there's sustainability."*
>
> - Executive Officer

The Project Manager is strongly convinced that collaboration is the most powerful approach when addressing major social challenges, while also acknowledging that it comes with difficulties due to inherent differences.

> *"It is my firm belief that we make progress more quickly when working together."*
>
> - Project Manager

4.2 Differences Between the Sectors

According to the Program Leader, some inherent differences lead to immediate clashes, often giving the collaboration a challenging start.

> *"It is very much about coming from different work environments, bringing different languages and workplace cultures, so a clash happened almost immediately."*
>
> - Program Leader

The Senior Advisor Business Sector elaborates on this by highlighting several attributes related to the cultural clash. Different visions, goals, and terminology are all causes of conflict. The Senior Advisor Business Sector further states that the concept of silo thinking is another relevant factor contributing to these clashes.

> *"So you can see that it's essentially about people thinking in silos, and people are extremely comfortable staying within those silos."*
>
> - Senior Advisor Business Sector

According to the Chief Strategy Officer, the inherent differences can be seen as almost ideological when the Private Sector engages in collaboration with Civil Society, especially in relation to NGOs. The remaining respondents, representing either Civil Society or the Private Sector, agreed on the challenge of reaching consensus between a profit-driven and aid-oriented mindset.

> *"Can't you understand that from our side, we are people centered and they're like, but you're also not understanding that from our side, we are profit centered."*
>
> - Executive Officer

The Chief Strategy Officer provides an example of this clashing mindset between profit and aid in relation to project planning. Further arguing that, in the initial phase of a project, NGOs are often heavily focused on implementingnumerous trainings and products without considering the market demand or needs. From a business perspective, this approach is seen as incomprehensible, as commercial logic would suggest a smaller initial implementation to assess the market's response.

When bringing the sectors together for the first time, the Program Leader argues that one is given a window of opportunity to prepare for the demanding climate that a Quadruple Helix setup can imply. These differences are often so fundamental that they need to be addressed upfront. The Program Leader further emphasizes that most actors involved in the collaboration share the same intention, although they come with different approaches.

”I myself, as the person in charge, have improved the way I introduce things and I now give a sort of warning, if I may put it that way - that you’re about to encounter completely different work cultures, and you have to accept that, take it or leave it.”

- Program Leader

The Chief Strategy Officer, representing the Private Sector expressed frustration over the slower work pace of the other sectors. The respondents representing Civil Society pointed out that working with Academia and Government can sometimes delay the process. The Senior Advisor Business Sector sympathizes with the previous respondents’ statement regarding the slower decision-making processes within government. Even so, the Senior Advisor Business Sector acknowledges that the government is fulfilling its intended role.

The Program Leader, who has experience in both the Private Sector and NGOs, expresses understanding of both perspectives. While recognizing the logic of the Private Sector, it is difficult to expect quick results when addressing complex issues. Nonetheless, the Program Leader emphasizes the value of learning from the private sector’s strengths to identify smaller targets or intermediate goals, as this can still provide valuable benefits.

”I know that it is naïve to expect quick results in certain areas. For example, when it comes to women’s rights, increasing women’s education or reducing domestic violence.”

- Program Leader

The Chief Strategy Officer pointed out that Academia and Civil Society may pause projects while waiting for donor funds, whereas the Private Sector tends to continue working regardless. According to the Executive Officer, many of these structural differences can become apparent early in a project, leading to extended discussions and negotiations before implementation even begins.

”The private sector is always moving forward, and doesn’t take breaks just because you’re waiting for the possibility of receiving project funding.”

- Chief Strategy Officer

4.3 Communication

All respondents mentioned communication as a key aspect of successful collaboration. The Program Leader highlights the use of internal language as critical, as it often precedes arising barriers and conflicts. The Project Manager adds that the use of internal language at times happens unintentionally, as a result of being used to exist and work in silos.

The Country Director highlights the importance of understanding that there are different languages used in profit-driven and aid-focused work. The Program Leader elaborates on this giving an example of how the Private Sector talks about 'delivering to the market', while NGOs refer to the same target group as 'right holders' rather than using market-related terms.

The Executive Officer mentions that sectors sometimes use different words to talk about the same thing. Still, it is important to remember, the Executive Officer continues, that what may not sound insulting in one context or sector can be perceived as insulting in another.

> *"Like, okay, you can't call them 'poor people'. You can't call them 'poor communities'. You can't call them 'people with no knowledge'."*
>
> - Executive Officer

The Chief Strategy Officer noted that some people from NGOs lack knowledge of basic entrepreneurial concepts, which affects the decision-making processes. According to the respondent, fundamental terms like 'return on investment' can be unfamiliar and cause confusion.

The Senior Advisor Business Sector highlights that buzzwords and acronyms should be used with caution. Explaining that, for example, many management-related acronyms are unfamiliar to those working in Civil Society, and people may feel uncomfortable admitting when they don't understand them. Talking from experience, the Senior Advisor Business Sector explains that this can lead to conversations where participants talk past one another instead of reaching mutual understanding.

The Researcher/ Consultant highlights the importance of "packaging the information", arguing that if the goal is to gain "buy-in" from all sectors, the project must be described in a way that clearly shows its value to each one. In other words, it's about speaking their language.

> *"So you really need to communicate the information package so that everybody sees not only organizational value but individual value as coming from different pillars."*
>
> - Researcher/ Consultant

4.4 Building Bridges

The Program Leader emphasizes common ground as an important factor in bridging gaps, elaborating that the focus should be on similarities and shared goals rather than on the differences that separate the parties. The Program Leader also argues that building bridges takes time and is an ongoing process that must be nurtured continuously.

The Country Director emphasizes the importance of knowing one's team. To fully engage team members and achieve the desired goals, it is essential to have a clear understanding of the individuals being led. According to the Program Leader, this also requires a solid understanding of teams and group dynamics, as well as trust among individuals.

> *"At the end of the day, it all comes down to people, and building trust between individuals."*
>
> - Program Leader

In addition, the Program Leader shares the insight that roles and responsibilities should be clearly articulated at an early stage. Roles should not be confused with responsibilities, and vice versa, the Program Leader argues. Based on experience, lack of a clear structure for roles and responsibilities can lead to unnecessary conflict and frustration within the group.

The Senior Advisor Business Sector emphasizes that each sector brings unique strengths to the collaboration, giving the example that the entrepreneurial drive is something the other sectors gain access to through this partnership.

> *"That's where you can bring in the entrepreneurial drive, willingness to take risks, the motivation to break new ground."*
>
> - Senior Advisor Business Sector

The Country Director makes a similar point in relation to co-creation, highlighting how their organization has learned the importance of data collection from working with Academia. For example, they now understand how statistics and proper documentation can strengthen their ability to influence policymakers.

> *"Yes, stories are good, but for policymakers, statistics is very, very helpful."*
>
> - Country Director

According to the Senior Advisor Business Sector, diplomacy is an essential element in bridging the cultural clashes in a Quadruple Helix setting. The Program Leader shares this view, stating that diplomacy is one of the crucial factors for making the collaboration successful.

”A bit of diplomacy is a prerequisite for a Quadruple Helix to function - and really, for any organization to function.”

- Senior Advisor Business Sector

The Executive Officer argues that strong leadership is essential to keep all parties calm and grounded during heated discussions, referring to past occasions where participants have raised their voices or even left the room. In such situations, the Executive Officer adds, it can be helpful to introduce new people to help the process move forward.

”In terms of like fixing such kind of conflicts, we have really tried dialogue or at times even introducing new people into the project.”

- Executive Officer

The Senior Advisor Business Sector highlights the importance of having people in the team with broad experience, preferably from several sectors, to help bridge the gaps. The Program Leader sympathizes with this, stating that individuals with long-standing experience in relevant fields are essential for well-functioning collaboration.

”You need to have that experience yourself, having worked for a while, preferably in different workplaces, environments, and countries. Yes, and having dealt with challenges and accepted them. If you haven’t, then you’re simply the wrong person in the room.”

- Program Leader

According to the Program Leader, who has extensive experience in cross-sector collaboration, certain personal qualities such as respect, attentiveness, and curiosity about each other’s work are always valuable when working in a Quadruple Helix setting. The Senior Advisor Business Sector also mentions traits that are important for creating the right conditions for successful co-creation.

”It’s not easy, it requires patience, a strong long-term perspective, attentiveness, and respect for other people’s expertise.”

- Senior Advisor Business Sector

5 Analysis and Discussion

This section presents an analysis of the results along with a discussion. It also includes a method discussion at the end of the section.

5.1 Strengths and Challenges of the Collaboration

All of the respondents promoted Quadruple Helix as a promising framework, while also acknowledging the challenges inherent in the co-creating model. When the partnership functions well the respondents agree that there is potential for sustainable outcomes. Sustainable in this context refers to the projects being anchored in multiple stakeholders sharing the same vision, which correlates well with the theory of innovation and value creation (Castaneda & Cuellar, 2020; Parveen et al., 2015).

All respondents agreed on the potential of synergy in the co-creating model. A contrasting perspective was given by the Chief Strategy Officer, who expressed skepticism toward the current structure of the framework, referring to significant knowledge gaps between sectors. If these gaps become too wide, the Chief Strategy Officer argued, the opportunity for co-creation and synergy is diminished and the fundamental idea of the collaboration is compromised. The Chief Strategy Officer's experience is in line with the downside of the cross-functional teams stated by Engelsberger et al. (2023), which addresses issues of work methods and professional values as relevant causes for friction in the collaboration (Engelsberger et al., 2023).

Bringing together four different actors from diverse fields in a Quadruple Helix setup will inevitably introduce differences in approaches, values, internal goals, organizational structure, and terminology use. All of the respondents agreed that these elements are potential barriers for causing interruptions for an efficient and successful partnership. These elements are also highlighted in Engelsberger's research as barriers for collaboration (Engelsberger et al., 2023).

Several respondents described differences in core values and focus areas between sectors as the two most prominent factors that cause division within the collaboration. Although no previous research has highlighted these factors as particularly dominant, a possible explanation may be the composition of personality traits in the collaboration, or in the characteristics of the projects in which they were involved. Furthermore, these elements that cause friction are not divergent from previous research of Quadruple Helix, stating that differing cultural backgrounds, goals, roles etc. can create barriers in the collaboration (Dobers & Stier, 2018)

The empirical findings in this study revealed that most internal tensions occurred

between the Private Sector and NGOs i.e. Social Society. While NGOs are typically driven by goals related to social justice, human rights, and community well-being, the Private Sector tends to operate according to a profit-oriented logic. There are several possible explanations for why these two silo mindsets have not yet reached synergy. One reason could be the inherent polarization between profit and aid driven logic. Another hypothesis is that the three actors- Academia, Government and Private Sector have longer tradition of collaboration within Triple Helix, and have already developed a more established working relationship.

A frequently mentioned issue was the differing expectations regarding deadlines, work pace, and decision making. The Chief Strategy Officer from the Private Sector shared frustration over the other sectors' slower work pace. This sense of frustration is described in Dobers and Stier (2018), where the difference in timeline is understood as one prominent factor for causing frustration and friction within the collaboration (Dobers & Stier, 2018).

5.2 How Communication Affects the Collaboration

It is clear that communication is a critical factor affecting the collaboration. All respondents, regardless of sector, emphasized its central role in terms of fostering understanding, while also acknowledging it as a potential source of misunderstanding and conflict.

In some cases, different sectors have different terms while meaning the same thing, while at other times sectors can be insensitive and unmindful when using terminology to describe issues. This inconsistent use of terminology is noted in Engelsberger et al. (2023) as a cause for communication barriers (Engelsberger et al., 2023).

According to both the Program Leader and the Senior Advisor Business Sector, the Private Sector and Civil Society tend to use different terms to describe the same target group. The example given was the Private Sectors' use of the term ”market” while Civil Society rather speaks about ”right holders” when referring to the same target group. This indicates that even though both sectors may refer to the same group of people, the framing can lead to misunderstandings. The importance of adapting communication to the target group is highlighted in Dobers & Stier (2018) (Dobers & Stier, 2018).

The Executive Officer also described a situation where the Private Sector used the term ”poor people” in a project proposal when addressing the target group. This caused a strong reaction from the NGO, since they rather address the target group as ”right holders”. In such cases, it is possible that the choice of word can shift the focus away from the actual issue and create frustration, which may lead to friction in the collaboration.

The use of internal or sector-specific language was another communication barrier mentioned by some respondents. The Senior Advisor Business Sector brought up that acronyms can exclude and people can feel too insecure to ask for clarifications when not understanding a certain word or acronym. This connects with the idea of ”intermediaries” described by Dobers and Stier (2018),

5.3 Bridging Differences Between Sectors

A recurring theme was the importance of building bridges. Given the variation of backgrounds, experiences and areas of expertise, building bridges becomes a critical component for successful partnership and collaboration. The Program Leader emphasized the importance of discussing the different mindsets and cultural backgrounds at an early stage to reduce the risk of friction and conflict further along in the collaboration. This insight shared by the Program Leader correlates well with the findings of Dobers and Stier's research, stating that different cultural backgrounds and communication styles can cause barriers in a Quadruple Helix framework (Dobers & Stier, 2018).

The variation of cultural background and communication styles may be considered a natural consequence of the inherent differences between sectors. Therefore, striving for alignment in approaches and goals becomes a priority issue to address at an early stage.

To overcome the inherent cultural differences among sectors, a well-functioning team is crucial. As emphasized by all respondents, the team itself plays a crucial role in building a successful and thriving partnership. Regardless of the complexity and extent of cross-collaborative projects, trust will always be one of the key factors determining the course for the partnership. The issue of trust was brought up by the Program Leader, which was also mentioned as an important aspect in Engelsberger's research regarding teams with strong collaborative ability (Engelsberger et al., 2023).

In regards to diplomacy, strong leadership during negotiations was promoted as another crucial aspect by the Executive Officer.

The Researchers/ Consultant emphasized the importance of a shared vision for a thriving partnership. The vision is the unifying element in the collaboration and it is not until all involving parties see the value of being a part of the collaboration that full commitment and partnership takes place, the Researcher/ consultant argues. The importance of a shared vision aligns with conclusions in Dobers & Stiers (2018), who state that a shared goal is essential to reach result in a Quadruple Helix setting (Dobers & Stier, 2018).

5.4 Method Discussion

During the review of the material, it was sometimes difficult to clearly hear what the respondents were saying in the recordings due to poor internet connections, which caused unclear or completely missing audio segments. This was addressed by giving all respondents the opportunity to fill in any missing information afterwards.

Video interviews may hinder the development of a relationship between the interviewer and the respondent. (Bryman et al., 2024)

After conducting the first interview, the interview guide was revised to improve the quality of the questions. This may be considered a potential source of error, as the remaining interviews were conducted using the updated version.

When selecting respondents for the study, the only criterion was previous experience in Quadruple Helix-driven projects. The level of experience varied among the respondents, which may have affect the quality of the results. To increase the reliability of the results, additional interviews covering a broader range of projects could have been conducted.

The two respondents representing Civil Society both worked at an NGO. This may imply an oversimplified representation of the Civil Society sector as a whole, given that the sector is broader and more diverse. As a result, the findings related to Civil Society should not be generalized to the entire sector.

Civil Society is a broad sector, but in this study, both respondents representing it worked at an NGO. As a result, the study focuses more specifically on NGOs and the results may not apply to other parts of Civil Society.

None of the respondents in the study represented the government sector, but the perspective is considered to be indirectly captured through the other respondents' experiences of collaborating with governmental actors. However, including a representative from the government sector would likely have increased the quality and reliability of the study.

6 Conclusions

This section presents the conclusions of the study, recommendations for actors in Quadruple Helix, and suggestions for further studies.

The aim of this research was to explore the perceived strengths and challenges of collaboration within a Quadruple Helix setting. Particular emphasis was placed on the role of communication and how differences between sectors might be bridged.

The study confirms that there are many inherent differences among the sectors, which both encourage innovation but also create barriers to collaboration. Identified challenges include differences in values, working methods, and communication. Attitudes toward the co-creation framework also varied. The study showed that changing behavior when entering a co-creation project is difficult, and old patterns of silo thinking were evident at all stages in collaborative projects. Identified strengths include the diversity of expertise and experience. When collaboration functions effectively, this diversity creates momentum for both synergy and innovation.

The study also shows that communication plays an important role in cross-sectoral collaboration. The use of acronyms and buzzwords can lead to misunderstandings, especially when people do not feel comfortable asking for clarification. Each sector has its own internal language, which can sometimes be difficult for others to understand. There is a particular need for sensitivity in the use of terminology between the private sector and the NGOs (Civil Society), given the polarized mindsets rooted in profit- and aid-driven logics. If not handled thoughtfully, there is a potential risk of mistrust and disrespect entering the collaboration.

A fundamental element in bridging gaps is maintaining an optimistic mindset and confidence in the co-creating framework. In addition, broad experience across various professional contexts is another important factor. Such experience provides valuable insight, enriching the collaboration by contributing multiple perspectives, approaches, and a nuanced mindset.

6.1 Recommendations for Actors in the Quadruple Helix

Based on the findings of this study it is recommended to use terms and acronyms thoughtfully, and to be mindful of explaining such terminology to ensure that all actors share a common understanding. It is also important to recognize that certain words may be sensitive in other contexts, or when used in regards to other sectors.

Furthermore, a *willingness* to understand other sectors and to show interest

in their fields and areas of expertise can enhance collaboration and help build stronger cross-sectoral relationships.

6.2 Suggestions for Further Studies

For further studies on cross-sectoral collaboration within the Quadruple Helix framework, it would be valuable to conduct a broader study with more respondents to capture a wider range of perspectives. Additional areas for further studies are listed below:

- A more focused study on the collaboration between the Civil Society and the Private Sector, as this study indicates that the interaction between these actors presents specific challenges.
- Further exploration of the Government's perspective on collaboration within the Quadruple Helix framework, as this actor was underrepresented in this study.
- A more specific study on how intercultural communication and competence are used in cross-sectoral settings and how they influence collaboration outcomes.

References

Birukou, A., Blanzieri, E., Giorgini, P., & Giunchiglia, F. (2013). A formal definition of culture. *Springer Nature*, *6*. https://doi.org/10.1007/978-94-007-5574-1_1

Bryman, A., Bell, E., & Harley, B. (2024). *Företagsekonomiska forskningsmetoder* (4th ed.). Liber AB.

Carayannis, E. G., & Campbell, D. F. (2012). *Mode 3 knowledge production in quadruple helix innovation systems*. Springer.

Castaneda, D., & Cuellar, S. (2020). Knowledge sharing and innovation: A systematic review. *Knowledge and Process Management*, *27*(3), 159–173. https://doi.org/10.1002/kpm.1637

Daspit, J., Justice Tillman, C., Boyd, N. G., & Mckee, V. (2013). Cross-functional team effectiveness. *Team Performance Management: An International Journal*, *19*(1/2), 34–56. https://doi.org/10.1108/13527591311312088

Dobers, P., & Stier, J. (2018). Quadruple helix co-creation in ssh: Experiences, considerations, lessons learned in a pan-european study in 12 countries. *Book of Papers*. http://urn.kb.se/resolve?urn=urn:nbn:se:du-29142

Engelsberger, A., Cavanagh, J., Bartram, T., & Halvorsen, B. (2023). Maximizing team development for open innovation in digital product development: The role of collaborative hrm and relational leadership. *Personnel Review*, *53*(4), 1022–1038. https://doi.org/10.1108/PR-09-2022-0657

González-Martinez, P., García-Pérez-Se-Lema, D., Castillo-Vergara, M., & Bent Hansen, P. (2021). Systematic review of the literature on the concept of civil society in the quadruple helix framework. *Journal of Technology Management Innovation*, *16*(4), 85–95. https://doi.org/10.4067/S0718-27242021000400085

Hammer, M. R., Bennet, M. J., & Wiseman, R. (2003). Measuring intercultural sensitivity: The intercultural development inventory. *International Journal of Intercultural Relations*, *27*(4), 421–443. https://doi.org/10.1016/S0147-1767(03)00032-4

Holliday, A. (1999). Small cultures. *Applied Linguistics*, *20*(2), 237–264. https://doi.org/10.1093/applin/20.2.237

Kava, C. M., Parker, E. A., Baquero, B., Curry, S. J., Gilbert, P. A., & Sauder, M. (2018). Organizational culture and the adoption of anti-smoking initiatives at small to very small workplaces: An organizational level analysis. *Tobacco Prevention and Cessation*, *4*(39). https://doi.org/https://doi.org/10.18332/tpc/100403

Leung, K., Ang, S., & Ling Tan, M. (2014). Intercultural competence. *Annual Reviews*, *1*, 489–519. https://doi.org/10.1146/annurev-orgpsych-031413-091229

Leydesdorff, L., & Etzkowitz, H. (1998). Triple helix of innovation: Introduction. *Science and Public Policy*, *25*(6), 358–364. https://doi.org/10.1093/spp/25.6.358

Lindberg, M., Lindgren, M., & Packendorff, J. (2014). Quadruple helix as a way to bridge the gender gap in entrepreneurship: The case of an innovation system project in the baltic sea region. *Journal of the Knowledge Economy*, *5*(1), 94–113. https://doi.org/10.1007/s13132-012-0098-3

Lloyd, S., & Härtel, C. (2010). Intercultural competencies for culturally diverse work teams. *Journal of Managerial Psychology*, *25*(8), 845–875. https://doi.org/10.1108/02683941011089125

Nguyen, H. T., & Marques, P. (2022). The promise of living labs to the quadruple helix stakeholders: Exploring the sources of (dis)satisfaction. *European Planning Studies*, *30*(6), 1124–1143. https://doi.org/https://doi.org/10.1080/09654313.2021.1968798

Park, N. K., Jang, W., Thomas, E. L., & Smith, J. (2021). How to organize creative and innovative teams: Creative self-efficacy and innovative team performance. *Creativity Research Journal*, *33*(2), 168–179. https://doi.org/10.1080/10400419.2020.1842010

Parveen, S., Senin, A. A., & Umar, A. (2015). Organization culture and open innovation: A quadruple helix open innovation model approach. *International Journal of Economics and Financial Issues*, *5*(1S), 335–342. https://doaj.org/article/57479cea84ca478ab954453c8d61940b

Popescu, A.-D., Suciu, S., & Raoult, M.-G. (2014). Intercultural competencies in collaborative teams. *Procedia Technology*, *12*, 577–582. https://doi.org/10.1016/j.protcy.2013.12.532

Schütz, F., Heidingsfelder, M., & Schraudner, M. (2019). Co-shaping the future in quadruple helix innovation systems: Uncovering public preferences toward participatory research and innovation. *She Ji: The Journal of Design, Economics, and Innovation*, *5*(2), 128–146. https://doi.org/10.1016/j.sheji.2019.04.002

Scott, C. (2023). *The project in international development: Theory and practice* (1st ed.). Routledge. https://doi.org/10.4324/9780429427411

Ulrich, K. T., & Eppinger, S. D. (2014). *Produktutveckling: Konstruktion och design*. Studentlitteratur.

United Nations. (2015b). *17 partnership for the goals* [Accessed 10 April 2025]. https://www.globalgoals.org/goals/17-partnerships-for-the-goals/

United Nations. (2015a). *Transforming our world: The 2030 agenda for sustainable development* [Accessed 10 April 2025]. https://sdgs.un.org/2030agenda

United Nations Children's Fund (UNICEF). (n.d.). *Goal 17: Partnerships for the goals* [Accessed 10 April 2025]. https://data.unicef.org/sdgs/goal-17-partnerships-for-the-goals/

Van Horne, C., & Dutot, V. (2017). Challenges in technology transfer: An actor perspective in a quadruple helix environment. *The Journal of Technology Transfer*, *42*(6), 285–301. https://doi.org/10.1007/s10961-016-9503-6

Wynn, D. C., & Clarkson, J. P. (2024). *The design and development process*. Springer.

Appendix A - Interview Guide in Swedish

Respondentens namn: **Sektor:** **Datum:**

INTRODUKTION
Presentera oss själva och förklara vem som gör vad under intervjun.

SYFTET
Syftet med denna studie är att undersöka de upplevda utmaningarna och styrkorna i samarbeten inom projekt som bedrivs inom Quadruple Helix. Fokus ligger på att analysera hur skillnader i organisationskultur påverkar tvärsektoriellt samarbete.

FÖRTYDLIGANDE
Innan vi börjar vill vi förtydliga att vår studie inte syftar till att utvärdera något specifikt projekt. Du är välkommen att dela exempel från tidigare projekt men vårt huvudsakliga fokus är att utforska dina bredare tankar och insikter om sektoröverskridande samarbete.

FÖRE INTERVJUN

- Samtycker du till att intervjun spelas in? (Materialet kommer endast vara tillgängligt för oss och raderas efter att projektet är avslutat.)
- Materialet kan skickas till dig i efterhand om du önskar, så att du får möjlighet att lägga till eller ändra innehåll.
- Har du några frågor innan vi börjar?

A) BAKGRUND OCH ERFARENHET

1. Kan du ge en kort introduktion om dig själv?
2. Kan du beskriva ditt nuvarande arbete och organisationen du arbetar inom?
   • Hur länge har du arbetat inom denna sektor?
3. Vad har du för erfarenhet av sektoröverskridande samarbeten?
   • Hur ofta samarbetar du med organisationer utanför din sektor?
   • Har du erfarenhet av att arbeta i en annan sektor?

B) UPPLEVDA UTMANINGAR I SAMARBETE

4. Vilka utmaningar kopplade till samarbete upplever du att det finns när olika sektorer arbetar med ett gemensamt projekt?
   • Har du själv stött på någon av dessa utmaningar? Kan du ge exempel?

5. Vilka utmaningar kopplade till språk och kommunikation upplever du att det finns?
   • Har du själv stött på någon av dessa utmaningar? Kan du ge exempel?
6. Vilka utmaningar relaterade till skillnader i organisationskultur upplever du att det finns?
   • Skillnader i normer, mål och intressen mellan sektorer och hur dessa påverkar samarbetet
7. Hur skulle du beskriva din förståelse för de olika sektorerna?
   • Förståelse för normer, intressen, mål etc.
   • Hur har du fått den förståelsen? Erfarenhet eller andra sätt?
8. Hur har samarbetsutmaningar hanterats i de sektoröverskridande projekt du har varit involverad i?
   • Vad fungerade? Vad fungerade inte?
   • Fick du det stöd du behövde? Hur? Varför? Vad saknades?

C) LÄRDOMAR OCH FRAMTIDA FÖRBÄTTRINGAR

10. Kan du berätta lite om vad du har lärt dig från att ha arbetat i sektoröverskridande projekt?
    • Vad har du lärt dig om samarbetet?
11. När du ser tillbaka, vad önskar du att du hade vetat i början av att arbeta i projekt över sektorsgränser?
    • Vilket råd hade du velat ge till någon som arbetar sektoröverskridande för första gången?
12. Vilka färdigheter och vilken kunskap anser du är avgörande för ett framgångsrikt samarbete över sektorer?
    • Varför? Exempel?
13. Vilka är de vanligaste konflikterna vid sektoröverskridande samarbeten?
    • Hur tror du att de skulle kunna undvikas?
14. Vilka förberedelser tror du är fördelaktiga?
    • Varför?

AVSLUTNING

15. Finns det något du skulle vilja lägga till?
16. Är det möjligt att kontakta dig senare om vi har några uppföljningsfrågor?
17. Som nämnts i början kan materialet skickas till dig i efterhand om du önskar, så att du får möjlighet att lägga till eller ändra innehåll.

Appendix B - Interview Guide in English

Respondent's Name: **Sector:** **Date:**

INTRODUCTION
Introduce ourselves and explain who will do what during the interview.

THE PURPOSE
The purpose of this study is to explore the perceived challenges and strengths of collaboration in projects conducted within the Quadruple Helix. The focus is on analyzing how differences in organizational culture affect cross-sector collaboration.

CLARIFICATION
Before we begin, we want to clarify that our study is not intended to evaluate any specific project. While you are welcome to share examples from your latest project, our primary focus is to explore your broader thoughts and insights on cross-sector collaboration.

BEFORE THE INTERVIEW

- Do you agree to have the interview recorded? (The material will only be available to us and will be deleted after the project is completed)
- The material can be sent to you afterwards if you wish, to give you the opportunity to add or modify the content.
- Do you have any questions before we start?

A) BACKGROUND AND EXPERIENCE

1. Can you give a brief introduction about yourself?
2. Can you describe your current work and the organization you're working in?
   • How long have you been in this sector?
3. What experiences do you have with cross-sector collaboration?
   • How often do you collaborate with organizations outside your sector?
   • Do you have experience working in different sectors?

B) PERCIEVED CHALLENGES IN COLLABORATION

4. What challenges do you believe exist when different sectors collaborate on a shared project?
   • Have you encountered any of these challenges yourself? Can you give any examples?

5. What challenges related to language and communication do you experience?
   • Have you encountered any of these challenges yourself? Can you give examples?
6. What challenges related to differences in organisational culture do you see?
   • Differences in norms, goals, interests between sectors and their effect on collaboration
7. How would you describe your understanding of different sectors?
   • Understanding of norms, interests, goals etc.
   • How did you gain this understanding? Through experience or other means?
8. How have collaboration challenges been handled in the cross-sector projects you have been involved in?
   • What worked? What didn't?
   • Did you recieve the support you needed? How? Why? What was missing?

C) LESSONS LEARNED AND FUTURE IMPROVEMENTS

10. Can you share a bit about what you have learned from working in cross-sector projects?
    • What have you learned about collaboration?
11. Looking back, what do you wish you had known in the beginning of working in projects across sectors?
    • What advice would you give to someone collaborating across sectors for the first time?
12. What skills and knowledge do you consider essential for a successful collaboration across sectors?
    • Why? Any examples?
13. What are the most common conflicts in cross-sector collaborations?
    • How do you think they could be avoided?
14. What preparations do you believe are beneficial?
    • Why?

CLOSING

15. Is there anything you would like to add?
16. Would it be possible to contact you later if we have follow-up questions?
17. As mentioned in the beginning, the material can be sent to you afterwards if you wish, to give you the opportunity to add or modify the content.

Appendix C - Quotes in Original Language

”Tror man inte på Quadruple Helix-modellen så är det ju dömt att misslyckas”
- Senior Advisor Business Sector **p.12**

”Min fulla övertygelse är att vi når fortare framåt tillsammans”
- Project Manager **p.12**

”Det handlar väldigt mycket om att man kommer från olika arbetsmiljöer, kommer med olika språk, olika arbetskulturer som det heter, så det blev krock ganska omedelbart.”
- Program Leader **p.13**

”Så ser ni att det egentligen handlar om att folk tänker i stuprör. Och folk är oerhört bekväma med att vara i stuprör.”
- Senior Advisor Business Sector **p.13**

”Jag själv, som ansvarig, har blivit bättre på introduktionen och varnar, om jag ska uttrycka mig så, att här kommer helt andra arbetskulturer och ni måste liksom acceptera det, take it or leave it”
- Program Leader **p.14**

”Jag vet att det är blåögt att förvänta sig snabba resultat inom vissa områden. T ex när det handlar om kvinnors rättigheter, öka kvinnors utbildningsgrad eller minska våld i hemmet.”
- Program Leader **p.14**

”Privata sektorn fortsätter ju alltid att köra på och tar ju liksom inte pauser för att du väntar på att kanske få in projektpengar.”
- Chief Strategy Officer **p.14**

”Till syvende och sist handlar det om människor, det handlar om bygga förtroende mellan människor.”
- Program Leader **p.16**

”Det är där man kan gifta in entreprenörernas drivkraft och möjlighet och riskbenägenhet och viljan att prova nya vägar.”
- Senior Advisor Business Sector **p.16**

”Lite diplomati är något som är en förutsättning för att en Quadruple Helix ska fungera, och vilken organisation som helst egentligen ska fungera.”
- Senior Advisor Business Sector **p.17**

”Man måste själv ha den erfarenheten, att man har jobbat en stund och jobbat helst på olika arbetsplatser, i olika miljöer, i olika länder. Ja, och tagit duster och accepterat det. Om man inte accepterat det, då är man ju liksom helt fel person i rummet.”
- Program Leader **p.17**

”Men det är svårt. Det kräver tålamod, det kräver väldigt mycket långsiktighet, det kräver lyhördhet och det kräver att man har respekt för de andra människornas kompetens.”
- Senior Advisor Business Sector **p.17**

TRITA – ITM-EX 2025:85
Stockholm, Sverige 2025

www.kth.se